

Speedup over PyTorch (1 GPU)

Strong Scaling

Weak Scaling

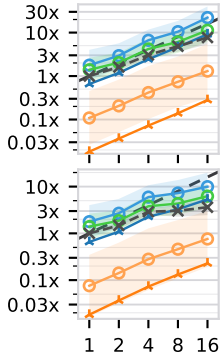
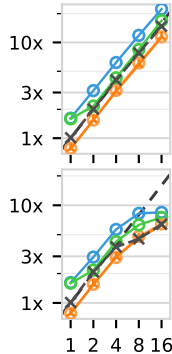
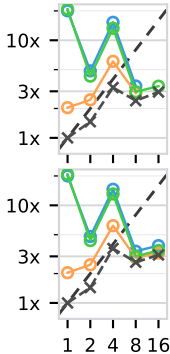
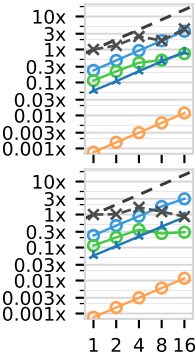
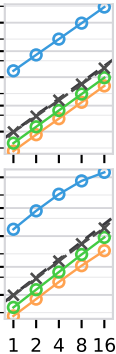
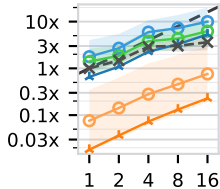
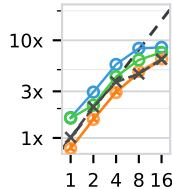
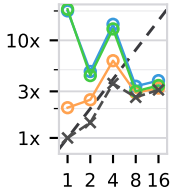
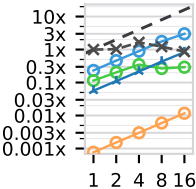
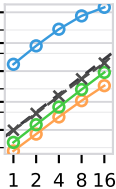
MoE Router

SDPA

All-to-All

Vocab Softmax

Geomean

GPUs P

-- Ideal Scaling of the PyTorch Baseline

—○— Qwen3.8-27B, No Skill Packet (n=15)

—○— OSS-120B, No Skill Packet (n=14)

—○— Kimi-K2.7, No Skill Packet (n=16)

—×— PyTorch Distributed (n=19)

—+— Qwen3.8-27B, RCCL Hints (n=5)

—+— OSS-120B, RCCL Hints (n=3)